

Co-designing AI-enhanced bilingual STEM e-books with families

Qualitative research | Family co-design + thematic analysis | 20 Latino families in co-design, 120 in RCT, children ages 5–7



Family-created design artifact

Parents' AI image created based on the story skeleton created during a co-design session

Research focus

How can bilingual e-books combine culturally familiar stories, AI-powered dialogue, and caregiver–child prompts to support early science learning and meaningful family interaction?

Approach

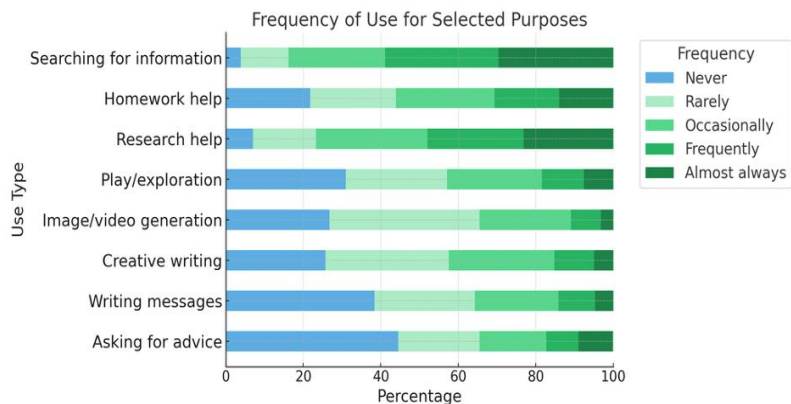
We conducted co-design sessions with latine families in Spanish focused on story content, language options, interactive features, and AI prompts. I analyzed transcripts, recordings, field notes, and family-created materials to translate families' experiences into design requirements.

Impact

- Established culturally relevant storytelling and flexible bilingual support as core family needs.
- Elevated digital safety and AI literacy as design priorities.
- Findings are shaping 24 bilingual e-books and an upcoming RCT evaluation.

Understanding student AI use in Mexico

Quantitative research | Survey design + statistical analysis | 7,500+ public high school students



Behavior across use cases

Frequency of generative AI use varied substantially by activity, revealing both adoption and literacy gaps. **Study Survey Question:** Have you used generative AI to help you with any of the following activities? (For each one, mark the frequency of use)

Research focus

How do adolescents use AI in their learning processes, and what individual, social, and contextual factors shape how and why they engage with it?

Approach

I contributed to a large-scale anonymous survey adapted from an established framework. Using quantitative and open-ended data, we examined AI use, confidence, self-concept, training experiences, perceived impact, and reliance.

Impact

- Translated findings from 7,500+ students into a policy brief with actionable guidance for educators, policymakers, and educational technology designers.
- Identified priorities for age-appropriate AI literacy, domain-specific teacher training, and AI tools that scaffold reasoning and self-regulation rather than replace student thinking.

See also [Policy Brief](#) & [ReVista article](#)

Understanding researcher needs for an AI-enabled scientific discovery tool

Qualitative research | In-depth interviews + workflow evaluation | Global, multidisciplinary research



Sanitized portfolio view

Note: The product interface, client identity, sample details, and proprietary findings are intentionally omitted.

Business issue

A technology team needed to understand how scientists generate and evaluate early-stage research ideas—and where an AI-enabled tool could provide meaningful support within complex, discipline-specific workflows.

Our solution

I conducted iterative qualitative research with scientists across multiple domains. In-depth interviews and workflow-based evaluations examined user needs, usability barriers, trust considerations, and the connection between idea generation and execution.

Impact

- Revealed meaningful differences in how scientists evaluate and develop research ideas.
- Identified opportunities to connect AI-assisted ideation with practical workflows.
- Findings informed product strategy and future experience improvements.

Evaluating solar workflows across European & LatAm markets

Qualitative research | In-depth interviews + workflow evaluation | Two European & three LatAm markets

Study Overview

Pre-use interview:
Current workflow
& needs

One week home
use: mid-week &
end-of-week
surveys

Post-use interview:
Experience &
workflow fit

Five markets • shared needs + local variation

Note: The client, product interface, participant counts, and proprietary recommendations are intentionally omitted.

Business issue

A technology team needed to understand how well a solar-data product supported professional workflows across global markets, where differences in data, regulation, financial practices, and project types could affect adoption.

Our solution

I conducted qualitative research with solar professionals in five markets, examining how the product fit into project-planning and decision-making workflows. I synthesized usability barriers, localized requirements, and unmet needs across key stages.

Impact

- Identified market-specific barriers affecting product relevance and usability.
- Distinguished localized requirements from needs shared across markets.
- Findings informed product priorities and international expansion opportunities.